

# Torsha Majumder

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## Professional Summary

### Lead ML Architect & Research Scientist (NVIDIA Academic Grant Recipient)

MS in Computer Science with a background in Computational Physics and 9+ years of professional experience in the USA and India. Spearheading NVIDIA-supported research to architect high-performance **SSL frameworks** for foundational-scale representation learning via **distributed Multi-GPU training (8x A100)**. Expert in architecting **multimodal Agentic Workflows (MAS)** and **DPO-aligned inference pipelines** (SageMaker/Docker). Specialized in bridging the gap between breakthrough scientific research and scalable industrial automation in high-stakes sectors.

## Core Technical Skills

- **ML & AI (Research):** Self-Supervised Learning (SSL) • Representation Learning • Vision & Time-series Transformers (DINOv2/ViT) • Responsible AI (Alignment/Safety) • DPO Alignment • Reinforcement Learning • SFT • PEFT • Knowledge Distillation (MiniLM/BERT) • Anomaly Detection • NLP (spaCy, NLTK)
- **Agentic AI & LLMs:** LangGraph • LangChain • Stateful Cyclic MAS • AI TRiSM (Guardrails) • RAG • Tool-Calling • Pydantic • Prompt Engineering • Gemini • Llama • OpenAI
- **Infrastructure & MLOps:** Distributed Training • Inference Optimization (ONNX) • AWS (SageMaker, S3, Boto3) • Docker • MLflow • VectorDB (Milvus, pgvector, Pinecone) • NoSQL (MongoDB) • Linux/Bash • JIRA/Confluence
- **Languages & Engineering:** Python • PyTorch • JAX • TensorFlow • NumPy • Pandas • Scikit-learn • SciPy • Matplotlib • SQL (PostgreSQL, Oracle) • JavaScript • React.js • Flask
- **Specialized Tools:** Hugging Face • PySpark • Dask • LSDB • Astropy • Git • LaTeX

## Professional & Research Experiences

### Nu Energy India | Lead ML Architect | Kolkata, India

Feb. 2026 – Present

- **Architected** a self-correcting **cyclic MAS (LangGraph/Gemini)** with **AI TRiSM guardrails** to automate industrial energy and safety auditing.
- Developing a **multimodal ingestion pipeline (Gemini Vision/Supabase)** for automated entity extraction from manuals and high-stakes instrument logs.
- Implementing a **Responsible AI framework** via a deterministic engine and **HITL verification**, reducing reporting from **weeks to days** with a **<0.5% error margin**.

### The SNAD Team | Lead ML Architect (NVIDIA Academic Grant Recipient)

Apr. 2024 – Present

- **Architected** **ASTRA**, a modular SSL framework; engineered **Astra-CLR**, a multi-filter time-series Transformer **trained on 8x NVIDIA A100 80GB GPUs** and scaled to generate embeddings for **~70M light curves**.
- **Engineered** a **multi-view late-fusion architecture** and **“multi-crop”** paradigm within ASTRA; utilized a **PEFT** strategy to achieve **77% fine-tuned accuracy**; research first-authored and **submitted to A&C Journal (arXiv)**.
- Spearheading the **ASTRA ecosystem expansion** by developing a breakthrough architectural advancement targeting **top-tier ML Conferences (2027)** for a **universal scientific foundation model** for photometric data.
- Discovered **8 Superluminous Supernovae (SLSN)** candidates in the ZTF DR8 release via an **Active Learning pipeline** utilizing the **"Pineforest" algorithm**, resulting in a first-author publication in **PoS**.

### Verizon | Data Scientist | Sunnyvale, CA, USA

Jun. 2021 – May 2023

- **Engineered** a scalable vendor recommendation and forecasting engine on **AWS SageMaker**, increasing prediction accuracy by **25%** through **collaborative filtering** and rigorous **A/B testing**.
- **Architected** an end-to-end automation pipeline for document classification and metadata extraction (**NER**) using **ResNet** and **OCR**, utilizing **MongoDB** and **Boto3** for **S3** integration; reducing manual operational effort by **15%**.
- **Led a 5-person cross-functional analytics team** for the FUZE Regulatory platform, overseeing the delivery of data-driven solutions to optimize nationwide business outcomes.

### MIT | Research Affiliate | Cambridge, MA, USA

Oct. 2021 – Nov. 2023

- **Optimized the DASH spectral classification engine**, migrating the architecture from **TensorFlow** to **Keras** and implementing **batch processing** to support novel transient classes (**Kilonovae, TDEs, AGNs**).
- **Engineered unsupervised ML pipelines** for **anomaly detection** and noise removal across massive **TESS, Kepler, and PLAsTiCC** datasets to identify rare exoplanetary and transient signals.
- **Co-authored research** in astronomical deep learning published at **NeurIPS 2023**.

- Engineered a **probabilistic neural network** to optimize **MCMC solver initialization** for exoplanetary orbital modeling; second-author research submitted to the **American Astronomical Society (AAS)**.

## Selected AI Portfolio

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- Sentinel-Hub (Multi-Agent System):** Architected a **multimodal**, 4-agent autonomous **cyclic MAS (LangGraph/Gemini)** with **AI TRiSM guardrails** to orchestrate cross-domain security forensics; integrated vision, transactional, and log-based **microservices** into a **stateful, self-correcting reasoning graph**.
- DeepShield (Responsible AI):** Aligned a **DINOv2** backbone via a **DPO-based PEFT approach** to prioritize human-perceived risk and AI Safety; achieved **97.5% sensitivity** on high-fidelity fakes and a **50.6% reduction** in False Positives on authentic media.
- LogSentinel (Infrastructure Security):** Engineered a **zero-day anomaly detector** using **distilled BERT** and **Zilliz (Milvus)** for **low-latency semantic analysis** on high-throughput HDFS logs; utilized **SSL representations** to identify **out-of-distribution system behaviors** and ensure infrastructure resilience.
- Fin-Intelligence (Secure Finance):** Engineered an **SSL-based bottleneck autoencoder** for transaction anomaly detection utilizing **Supabase (pgvector)** and **HITL feedback**, and a **Temporal RAG** pipeline with **Pydantic-enforced validation (AI TRiSM)** to automate **multi-modal** parsing and comparative multi-quarter financial analysis.

## Education

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**MS in Computer Science (Thesis) | The University of Texas at Dallas, USA | 2018 – 2020**

*Thesis: Ensembles of Oblique Decision Trees (ML & Algorithm Design)*

**Graduate Research (Physics) | The University of Lethbridge, Canada | 2024 – 2025**

*Transition-Edge-Sensor (TES) optimization for Double Fourier Interferometry; Instrumentation Group.*

**B.Tech in Information Technology | Maulana Abul Kalam Azad University of Technology, India | 2013 – 2017**

*Project: Institute Library Management System (DBMS)*

## Publications

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- Majumder, T., et al. (In Prep). Large-Scale Self-Supervised Pre-training for Representational Learning in Time-Domain Astronomy.** Targeting *top-tier ML conferences (2027)*.
- Majumder, T., Malanchev, K., Ishida, E. E. O. Multi-Scale Contrastive Attention for Light-Curve Representation Learning.** Submitted to *Astronomy and Computing (A&C) Journal*, July 2026. [arXiv:2606.31627](#).
- Huang, H., ... **Majumder, T., et al. Predicting the Age of Astronomical Transients from Real-Time Multivariate Time Series.** *Neural Information Processing Systems (NeurIPS 2023)*. [arXiv:2311.17143](#).
- Majumder, T., et al. Superluminous Supernova Search with PineForest.** *Proceedings of Science (PoS)*, PoS(MULTIF2025)041. Accepted Jan 2026.
- Schussler, J., **Majumder, T.**, Penev, K. **Tidal Dissipation Leading to Circularization in Kepler Binaries.** Submitted to the *American Astronomical Society (AAS) Journal*, Apr 2026.
- Majumder, T. (2020). Ensembles of Oblique Decision Trees.** *UTD Master's Thesis*. URI: [hdl.handle.net/10735.1/8818](http://hdl.handle.net/10735.1/8818).

## Awards & Honors

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- NVIDIA Academic Grant Recipient | NVIDIA | Oct. 2025 – Mar. 2026**
- Graduate Research Award (ULGRA) | The University of Lethbridge | 2024**
- Associate Member (AMIE) | Institute of Engineers (India), Computer Science & Engineering | 2017**
- Best Final-Year Project Award | Institute Library Management System, STCET | 2017**

## Talks & Mentorships

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- Selected Talks:** *MIT TESS Science Talk*, Cambridge, MA (2022); *Int'l Meeting on Eclipsing Binaries*, China/Online (2024)
- Mentorship:** Led **Data Analytics & FinTech bootcamp programs** at 2U Inc./edX for US/Australian universities (2020 – 2023).
- Instruction:** **Graduate Teaching Assistant** for Biophysics and Engineering Mechanics at The University of Lethbridge (2024 – 2025).